The Fermionic Matter Sub-Programme

Presentation Note 6

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Abstract

The fermionic matter sub-programme of the Cosmochrony corpus addresses a single central question: where do fermions, chirality, hypercharge, and three generations come from? The answer is that they are not postulated as external fields: they arise as the spinorial face of the admissible Weil module V_ρ, once its metaplectic Lie-algebraic structure is complexified by the Lorentzian metric established in the geometric branch. The structural chain is:

$$\Pi \Rightarrow F_n \simeq V_\rho \Rightarrow \mathrm{Mp}(2,\mathbb{R}) \Rightarrow \mathrm{mp}(2,\mathbb{R})_\mathbb{C} \simeq \mathfrak{sl}_2(\mathbb{C}) \rightsquigarrow \mathrm{Spin}(3,1) \simeq \mathrm{SL}(2,\mathbb{C}) \Rightarrow \mathcal{S}_\Pi.$$

Three modular structural results are established in Q14: (A) the admissible spinor bundle $\mathcal{S}_\Pi$ reproduces the $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ electroweak bundle structure from tensor functors of $\mathcal{S}_\Pi$; (B) the projected Dirac operator $D_{\Pi,g,A}$ contains a canonical projective endomorphism E_Π that enforces the $V - A$ chiral structure (established via the spinorial BI lift theorem of Q14) and whose γ_5-weighted a_4 coefficient imposes anomaly-cancellation constraints making hypercharge a spectral datum; (C) the saturation invariant $\sigma_c(n_3) = 3$ has a spinorial multiplicity reading yielding a gauge-singlet three-generation factor C^3_{gen}. The colour-coupled quark sector is unconditional at the pointwise level ($[\text{H-color}]_{\text{pointwise}}$ established in O31). The structural form of E_Π is now fixed as a Schur complement of the eliminated spinorial block, and the generation-split amplitude mechanism is derived as Born–Infeld saturation in the companion note [8]; the open deliverables are its explicit Lorentzian block and the Yukawa sector, while the split value itself is dictionary-bound, its first-principles derivation relocated upstream to the ADE case selection and the cascade exponent.

Keywords. fermionic matter; admissible spinor bundle; metaplectic lift; chirality; V-A structure; hypercharge rigidity; anomaly cancellation; three generations; projected Dirac operator; Weil representation; non-injective projection; presentation note.

Contents

1 Motivation and Central Question

The fermionic matter sub-programme answers the following question.

Central question. The bosonic spectral stratification — gravity at a_2, Yang–Mills at a_4 — derives the bosonic sector of the Standard Model from the same admissible spectral functional. Fermions, chirality, hypercharge assignments, and the three-generation structure are not covered by the bosonic synthesis. *Do these fermionic structures also arise as forced consequences of the admissible Weil fibre, or must they be introduced as additional postulates?*

The answer established in Q14 [5] is: they are forced. Fermions arise as the spinorial face of the Weil module V_ρ after Lorentzian complexification of its metaplectic lift. Chirality and the $V - A$ structure arise from the projective endomorphism E_Π of the projected Dirac operator, via the spinorial lift of BI parity established in Q14. Hypercharge is not a free parameter: it is constrained by spectral anomaly cancellation. Three generations arise from the spinorial multiplicity reading of the same rank-three invariant $\sigma_c(n_3) = 3$ that yields the three spatial directions in the geometric branch.

This sub-programme is the first paper of the Q-series that depends simultaneously on all four layers of the programme: the admissible Weil fibre structure (Notes 1 and 5), the closed Lorentzian geometry (Note 2), the gauge bundle (Note 3), and the gauge–gravity synthesis (Q13 [6], to be summarised in Presentation Note 8).

Remark 1 (Scope: identification only, not mass spectrum)**.** This sub-programme closes the *identification* of the fermionic sector: spinor bundle, chirality, hypercharge quantum numbers, and generation factor. It does not derive the mass spectrum, the Yukawa couplings, or the CKM/PMNS mixing matrices. Those require the explicit construction of E_Π as an operator on the spectral space, and the integration of the projected Yukawa Lagrangian $\mathcal{L}_Y$, both of which are listed as open deliverables.

Remark 2 (Vocabulary note)**.** Throughout this note, *admissibility constraint* replaces the earlier "relaxation" vocabulary. The substrate χ is static; fermions are not excitations of χ but the spinorial interpretation of the admissible fibre after geometric complexification.

2 Position in the Cosmochrony Programme

The fermionic matter sub-programme sits at the apex of Branch III. It depends on all five preceding sub-programmes and provides the fermionic matter content consumed by any downstream phenomenological application.

Inputs from upstream:

- $F_n \simeq V_\rho \simeq L^2(\mathbb{Z}/q\mathbb{Z})$, canonical symplectic structure of V_ρ, and the parity involution $c \leftrightarrow q - c$ (Notes 1 and 5).
- Effective Lorentzian metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ and effective base manifold $M_{\text{eff}} = \mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$ (Note 2, Q5b–Q11).
- Admissible principal bundle $P_{G_\Pi}(M_{\text{eff}}, G_\Pi)$ and gauge connection A (Note 3, Q6a).
- $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ gauge group (Note 3).
- Spectral entropy functional $S_\Pi[g, A]$ and the a_4 variation (Note 4 [11]/Q12, Q13 [6]).
- $H_{\text{eff}} = \mathrm{Sym}^2(V_\rho)$, three spatial directions from $\mathrm{Im}\,\mathbb{H}$, $\sigma_c(n_3) = 3$ (Note 2, Q7 [12]).

3 Logical Chain of the Sub-Programme

The sub-programme is organised around the following structural chain.

$$\Pi \Rightarrow F_n \simeq V_\rho \;\Rightarrow\; \mathrm{Mp}(2,\mathbb{R}) \;\Rightarrow\; \mathrm{mp}(2,\mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C}) \;\rightsquigarrow\; \mathrm{Spin}(3,1) \simeq \mathrm{SL}(2,\mathbb{C}) \;\Rightarrow\; \mathcal{S}_\Pi \tag{1}$$
$$\Rightarrow \underbrace{\mathrm{Sym}^2(\mathcal{S}_\Pi) \Rightarrow \mathrm{ad}_{\mathbb{C}}(P_\Pi)}_{\mathrm{SU}(2)_L},\quad \underbrace{\wedge^2(\mathcal{S}_\Pi) = L_Y}_{\mathrm{U}(1)_Y},\quad \underbrace{E_\Pi \text{ left-admissible}}_{V-A},\quad \underbrace{\sigma_c(n_3) = 3 \to C^3_{\mathrm{gen}}}_{3\text{ generations}}.$$

Three distinct mechanisms contribute.

1. *Lorentzian complexification forces the spin group.* The Weil module V_ρ carries a canonical symplectic structure; its automorphisms lift to $\mathrm{Mp}(2,\mathbb{R})$. The effective metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ (Lorentzian) forces the Lie algebra complexification $\mathrm{mp}(2,\mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C})$, whose Lorentzian integration is $\mathrm{Spin}(3,1) \simeq \mathrm{SL}(2,\mathbb{C})$. The spinor bundle $\mathcal{S}_\Pi$ is the associated fundamental bundle.

2. *Tensor functors of $\mathcal{S}_\Pi$ reproduce the electroweak bundle.* $\mathrm{Sym}^2(\mathbb{C}^2) \simeq \mathfrak{sl}_2(\mathbb{C})$: the symmetric sector gives the complex adjoint, from which the compact real form selects the $\mathrm{SU}(2)_L$ gauge sector. $\wedge^2(\mathbb{C}^2) \simeq \det(\mathbb{C}^2)$: the determinant line L_Y supports the $\mathrm{U}(1)_Y$ factor. No independent electroweak matter bundle is introduced.

3. *The projective endomorphism E_Π enforces chirality and hypercharge.* The projected Dirac operator satisfies the Lichnerowicz-type identity
$$D^2_{\Pi,g,A} = -(\nabla^{S,A})^2 + \frac{R}{4} + \frac{1}{2}\gamma^\mu\gamma^\nu F^a_{\mu\nu} T^a_R + E_\Pi.$$
By the spinorial lift of BI parity, established in Q14 [5], E_Π is left-admissible ($P_R E_\Pi P_R = 0$), giving the $V - A$ structure. $\mathrm{U}(1)_Y$ invariance of $S_\Pi[g, A, D_{\Pi,g,A}]$ imposes the anomaly-cancellation trace constraint $\mathrm{Tr}_{\mathcal{S}_\Pi}(\gamma_5 Y A_\Pi(x)) = 0$, fixing hypercharge as a spectral coherence datum.

4 Constituent Papers

The sub-programme has a single primary paper.

4.1 Q14: Fermionic matter and chirality from projective Dirac admissibility

Q14 [5] is the sole constituent paper. It is organised around three independent but mutually compatible structural results.

Theorem A: Spinorial electroweak bundle (structural; unconditional). The admissible Weil module V_ρ induces, via the metaplectic lift and the Lorentzian complexification forced by $g^{\mu\nu} = 2\eta^{\mu\nu}$, an admissible spinor bundle $\mathcal{S}_\Pi$. Its tensor sectors satisfy:

$$\mathrm{Sym}^2(\mathcal{S}_\Pi) \simeq \mathrm{ad}_{\mathbb{C}}(P_\Pi), \qquad \wedge^2(\mathcal{S}_\Pi) = L_Y.$$

After selection of the compact real form, these define the $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ electroweak bundle data without any additional input.

Theorem B: Chiral selection and hypercharge rigidity (structural; unconditional). The projected Dirac operator $D_{\Pi,g,A}$ has the Lichnerowicz-type square above. By the spinorial lift of BI parity, established in Q14 (the antilinear $J_\Pi = HK$ maps right-chiral modes to non-admissible directions), E_Π is left-admissible: $P_R E_\Pi P_R = 0$. This is the $V - A$ structure. $\mathrm{U}(1)_Y$ invariance of the projected spectral functional then imposes, at the a_4 level:

$$\mathrm{Tr}_{\mathcal{S}_\Pi}(\gamma_5 Y A_\Pi(x)) = 0 \quad \forall x \in M_{\mathrm{eff}},$$

where A_Π is the γ_5-weighted a_4 Seeley–DeWitt density of $D^2_{\Pi,g,A}$. This constraint is not postulated: it is the chiral consistency condition of the same a_4 level that yields Yang–Mills dynamics in the bosonic variation. The hypercharge weights Y_R are thereby not free parameters but spectral coherence data.

Theorem C: Generation multiplicity (structural; unconditional). The saturation invariant $\sigma_c(n_3) = 3$ has two functorially distinct readings of the same rank-three admissible invariant. In the geometric branch (Note 2), it yields three effective spatial directions. In the spinorial branch, the spinorial multiplicity functor maps it to a gauge-singlet generation space:

$$C^3_{\mathrm{gen}} \subset \ker(\mathrm{ad}_{\mathrm{SU}(2)} \oplus Y).$$

The identity $\mathbb{R}^3_{\mathrm{space}} \neq C^3_{\mathrm{gen}}$ is established explicitly (Q14 §5): the spatial and generation spaces are distinct functorial images of the same admissible invariant. Three generations are forced as a gauge-singlet factor. A qualitative mechanism for their splitting is identified in Q14 §6 (summarised below); the explicit E_Π and the amplitude of the splitting remain open.

Dynamic generation lifting (localisation and A4 matching). The internal structure of C^3_{gen} is probed by the spectrum of E^2_Π (Q14 §6). A static J_Π-real, weight-preserving restriction of E^2_Π has a degenerate outer pair: the two light generations cannot be split statically (static obstruction). The admissible lifts removing this degeneracy form a two-dimensional J_Π-odd sector, the real-split direction being the weight generator J_3 and the mixing direction a doublet rotation. The generation split is not a finite-front observable in the symmetric sense: the raw finite front remains J_Π-equivariant and gives $u_{\mathrm{fin}} = 0$ [10]. After Lorentzian complexification, however, the chiral projector makes it meaningful to read a J_Π-odd frontier cocycle, and the residual automorphism is the central Weil lift $R_b = W(-I) = \mathrm{FT}^2$, which commutes with γ_5: it is blind to chirality and does not obstruct the $V - A$ selection [9]. The remaining amplitude is fixed at A4 by matching the discrete chiral-frontier normalisation $\mathcal{N}_A$ [7] with the Lorentzian Born–Infeld curvature $\mu^2_\chi = \partial^2_s \mathcal{B}(0)$ [10], neither ingredient reducing to the other. Generation mixing further requires the complex metaplectic phase.

Colour sector (unconditional at the pointwise level). The colour-coupled quark sector $\mathcal{S}_\Pi \otimes V_{\mathrm{color}}$ is obtained by tensoring the admissible spinor bundle with the colour module V_{color}. The status of this sector inherits the status of [H-color] from Notes 1 and 3: established unconditionally on the colour-adapted graph, and on the standard graph at the rank, averaged, and effective-exponent levels, with [H-color]$_{\mathrm{pointwise}}$ proved analytically (O31, exact equality of BFS capacity profiles at finite q). The only residual component is the extension beyond the standard Cayley graph to general generating sets, which is non-blocking for the quark bundle.

Status of the spinorial BI lift. The spinorial lift of BI parity to the spinor bundle $\mathcal{S}_\Pi$ is established as a theorem in Q14: the BI parity involution $c \leftrightarrow q - c$, proved at the scalar Weil level in O18 and BornInfeld, lifts to the antilinear endomorphism $J_\Pi = HK$ satisfying $J^2_\Pi = -1$ and $J_\Pi \gamma_5 = -\gamma_5 J_\Pi$, dressing the parity by the weight grading $H = 2J_3$. Chirality and $V - A$ in Q14 are therefore unconditional, their only residual dependence being the axiomatic chain A1–A4 [1] and the upstream BI parity datum (O30, U1).

4.2 Angular Amplitude Reduction: the J_3 split as oriented symplectic area [7]

This note fixes the numerical observable for the absolute normalisation of the angular generation split before any Weil–BFS computation. A Baker–Campbell–Hausdorff reduction identifies the per-step J_Π-odd J_3 component of the metaplectic cascade with the oriented symplectic area swept per step — the central, Carnot-degree-2 increment of Heis_3 — and not with a free shear

Table 1: Summary of Q14 results by theorem. Status: P = proved; S = structural.

Result	Central output	Status
Theorem A	Spinor bundle $\mathcal{S}_\Pi$; EW bundle from tensor functors	S (uncond.)
Theorem B ($V-A$)	$P_R E_\Pi P_R = 0$; $V-A$ structure	S (uncond.)
Theorem B (hypercharge)	Y_R from anomaly-cancellation trace; not free	S (uncond.)
Theorem C	C^3_{gen}; three generations as gauge-singlet factor	S (uncond.)
Colour sector	$\mathcal{S}_\Pi \otimes V_{\text{color}}$; quark bundle	S (uncond., pointwise)

parameter. The quantity to measure is the accumulated oriented J_3-odd area along the cascade, normalised by the projected capacity. A direct evaluation on symmetric BFS shells returns $\Theta_{\text{raw}} = 0$ identically: the symmetric capacity data fix the radial sector but cannot select an oriented angular branch, so the orientation prescription is an irreducible input prior to the normalisation $\mathcal{N}_A$.

4.3 Q11 Oriented Frontier: recursive diagnostic and resolution of the operator obstruction [9]

The symmetric-shell observable vanishes ($\Theta_{\text{raw}} = 0$) because the involution $g \mapsto g^{-1}$ pairs opposite central phases; this note defines the honest replacement, a frontier transfer observable on the directed edges $g \to gs$ whose central increment is the Heisenberg cocycle $a(g)\, s_b$, the discrete counterpart of $\frac{1}{2}[X, Y]$ and hence of the dynamical J_3. Its main result resolves the operator obstruction: the residual reflection $R_b = \varphi \circ J_\Pi^{-1} = W(-I) = \mathrm{FT}^2$ is the central Weil lift, and $[\gamma_5, R_b] = 0$, so R_b is blind to chirality and the $V-A$ selector removes the φ-paired obstruction without a spacetime parity. What remains is the amplitude normalisation $\mathcal{N}_A$, not the operator question.

4.4 Projective Residue as a Schur Complement: localisation and finite/Lorentzian separation [10]

This note fixes the structural status of the split u before the explicit E_Π. The projected Dirac square has the Schur–Feshbach form $E_\Pi = -M^\dagger M$ with $M = (1-P)\, D\, \Pi_S^*$, exhibiting E_Π as the Schur complement of the eliminated spinorial block; u is localised in the J_Π-odd part of $E_\Pi^2|_{C^3_{\text{gen}}}$ as the antiunitary equivariance defect $\Delta_\chi(P) = \pi_{LL} - \tau\, \overline{\pi_{RR}}\, \tau^{-1}$. The finite locking is J_Π-equivariant, so $u_{\text{fin}} = 0$ (finite/Lorentzian separation): the split cannot originate in the finite cascade and is a Lorentzian datum, its magnitude reduced to the Born–Infeld saturation curvature $\mu_\chi^2 = \partial_s^2 \mathcal{B}(0)$. The Schur transversality on which the A4 matching was conditional is reduced, in the same note, to the non-vanishing of the projected commutator $[\partial_s P|_0, J_\Pi]\big|_{\text{span}(e_+, e_-)}$, with an exhaustive transverse/null dichotomy including the zero-mode subcase. The determinantal Born–Infeld margin $\mathcal{B}(s)$, the antisymmetric structure of the chiral modulus, and the reduction of μ_χ^2 to the Lorentzian genus of the chiral polarisation are developed in the companion note [8]. That companion further derives the amplitude mechanism: on the corpus-derived real symplectic cascade the transverse (spin-two) cubic response vanishes path-invariantly at the complex metaplectic phase $\gamma = 0$, for every ordering and reference, so the derived mechanism is Born–Infeld saturation while the split persists ($u \neq 0$). The sixth-order interior lock is doubly conditional — on a non-derived complex phase and on a non-prescribed cascade ordering — and not concluded; the split value $|u|$ remains dictionary-bound through the chiral-frontier normalisation $\mathcal{N}_A$.

4.5 Arithmetic orthogonality of the ADE gate: the conditional status of ε [4]

This transversal note isolates the logical status of the generation-split value $\varepsilon = 1/10$. It is a theorem modulo the ADE case-selection gate: the algebra $\frac{1/2+\varepsilon}{1/2-\varepsilon} = \frac{3}{2} \Rightarrow \varepsilon = 1/10$ is trivial and case-independent, so the first-principles status of ε is exactly that of the gate. The obstruction to the gate is characterised as an arithmetic orthogonality. In the compositum $\mathbb{Q}(\zeta_q, \sqrt{5})$ the Born–Infeld parity $c \mapsto q - c$ is complex conjugation ($\zeta_q \mapsto \zeta_q^{-1}$, $\sqrt{5} \mapsto \sqrt{5}$), lying in a distinct direct factor of the Galois group from the Galois-spin action ($\zeta_q \mapsto \zeta_q$, $\sqrt{5} \mapsto -\sqrt{5}$) that selection of the $\sqrt{5}$-locus would require. The same fact places the order-five versus order-ten alternative below projective parity resolution, consistent with the central reflection $R_b = W(-I)$ being scalar on the chiral carrier [9]. The negative theorem — that A1–A4, ENI Corollary 6 and the A4 locking do not supply the Galois-spin action — is a no-go for the gate, unconditional in the present corpus construction by a character-table audit of $2I$; only the full-tower generalisation to every conceivable recursive stratum remains open. Hence $\varepsilon = 1/10$ is conditional on the gate rather than an independent dictionary parameter, and the first-principles front for the value is the gate itself: upstream and structural.

5 Inputs from Upstream Results

1. $F_n \simeq V_\rho$ **and symplectic structure** (Notes 1, 5): the identification of the admissible fibre as the Weil representation is the starting point of the metaplectic lift. Without this, the spinor bundle $\mathcal{S}_\Pi$ has no canonical definition.

2. **Lorentzian metric** $g^{\mu\nu} = 2\eta^{\mu\nu}$ (Note 2, Q5b–Q11): the Lorentzian signature forces the complexification $\mathrm{mp}(2,\mathbb{R})_\mathbb{C} \simeq \mathfrak{sl}_2(\mathbb{C})$. Without the Lorentzian metric, the spin group would remain $\mathrm{Sp}(2,\mathbb{R})$ and no Dirac spinors would arise.

3. **Gauge bundle and** $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y \times \mathrm{SU}(3)$ (Note 3, Q6a): the admissible principal bundle and gauge connection A enter the spin–gauge connection $\nabla^{S,A}$ of the projected Dirac operator.

4. a_4 **Yang–Mills variation** (Q12 [13], Q13 [6]): the spectral anomaly cancellation of Theorem B uses the same a_4-level structure that yields Yang–Mills dynamics in the bosonic sector. The fermionic a_4 coefficient is a structural extension of the bosonic one. The gauge–gravity synthesis of Q13 is the scientific dependency here; its presentation-level treatment will appear in Presentation Note 8.

5. $\sigma_c(n_3) = 3$ (Note 1, O23): the rank-three saturation invariant provides the spinorial multiplicity input for Theorem C.

6. **BI parity involution** (BornInfeld, O18): the scalar BI parity $c \leftrightarrow q - c$ is the input from which the spinorial BI lift is derived; its derivation at the scalar level is complete.

6 Outputs to Downstream Branches

7 Status of Results

7.1 Structural results (unconditional)

1. *Admissible spinor bundle* (Theorem A): $\mathcal{S}_\Pi$ is induced from V_ρ via the metaplectic lift and Lorentzian complexification. The construction is algebraic and functorial; it depends on the Lorentzian closure of Q5b–Q11 but not on the spinorial BI lift.

Table 2: Principal outputs and downstream consumers.

Output	Consumer	Status
Admissible spinor bundle $\mathcal{S}_\Pi$	Phenomenology, Q-series	S
$\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ from tensor functors	SM phenomenology	S
$V - A$ chiral structure	Weak interaction pheno	S (uncond.)
Hypercharge as spectral datum	SM charge assignment	S (uncond.)
Three-generation factor C^3_{gen}	Mass/mixing notes	S
Quark bundle $\mathcal{S}_\Pi \otimes V_{\mathrm{color}}$	QCD sector	S (uncond., pointwise)
E_Π as projective endomorphism	Yukawa sector (open)	S (open)

2. $\mathrm{SU}(2)_L$ *from* $\mathrm{Sym}^2(\mathcal{S}_\Pi)$ (Theorem A): the symmetric tensor sector gives the complex adjoint $\mathrm{ad}_{\mathbb{C}}(P_\Pi) \simeq \mathfrak{sl}_2(\mathbb{C})$; after compact real-form selection, the $\mathrm{SU}(2)_L$ gauge sector is identified.

3. $\mathrm{U}(1)_Y$ *from* $\wedge^2(\mathcal{S}_\Pi)$ (Theorem A): the determinant line $L_Y = \wedge^2(\mathcal{S}_\Pi)$ is the geometric support of the hypercharge factor.

4. *Generation multiplicity* (Theorem C): $C^3_{\mathrm{gen}} \subset \ker(\mathrm{ad}_{\mathrm{SU}(2)} \oplus Y)$ is a gauge-singlet three-generation space arising from the spinorial multiplicity reading of $\sigma_c(n_3) = 3$. The spatial and generation readings of the rank-three invariant are distinct ($\mathbb{R}^3_{\mathrm{space}} \neq C^3_{\mathrm{gen}}$; Q14 §5).

7.2 Further structural results (unconditional)

1. *Left-admissibility of* E_Π (Theorem B): $P_R E_\Pi P_R = 0$ follows from the spinorial BI lift (Q14), now established.

2. $V - A$ *chiral structure* (Theorem B): the condition $P_R E_\Pi P_R = 0$ selects the left-chiral admissible projection $P_L \mathcal{S}_\Pi$, giving the $V - A$ coupling of weak interactions.

3. *Hypercharge rigidity* (Theorem B): the anomaly-cancellation trace condition $\mathrm{Tr}_{\mathcal{S}_\Pi}(\gamma_5 Y A_\Pi(x)) = 0$ constrains Y_R to spectral coherence data; hypercharge is not a free parameter of the framework.

4. *Quark bundle* (§6): $\mathcal{S}_\Pi \otimes V_{\mathrm{color}}$ is structural and unconditional at the pointwise level: the spinorial BI lift is established (Q14) and $[\text{H-color}]_{\mathrm{pointwise}}$ is proved analytically (O31).

5. *A4 Schur-transversality reduction* [10]: the conditional residual of the A4 matching is reduced to the non-vanishing of the projected commutator $[\partial_s P|_0, J_\Pi]\big|_{\mathrm{span}(e_+,e_-)}$, with an exhaustive transverse/null dichotomy including the zero-mode subcase; the transverse branch determines the split via the electric genus of the chiral polarisation [8], the Schur-null branch is marginal.

7.3 Open problems

1. *Explicit form of* E_Π. E_Π is identified as the zero-order endomorphism of the projected Dirac square, the "internal spectral residue of non-injective projection." Its structural form is now fixed as the Schur complement of the eliminated spinorial block, $E_\Pi = -M^\dagger M$ with $M = (1 - P) D\, \Pi^*_S$ [10]; what is not yet constructed is its explicit Lorentzian block $1 - P(s)$ along the J_Π-odd modulus. The inter-generation splitting is localised in the J_Π-odd part of E^2_Π; its amplitude mechanism is the derived Born–Infeld saturation [8]. The

split value $\varepsilon = 1/10$ is not an independent dictionary parameter: it is forced algebraically once the ADE case-selection gate supplies the external $3:2$ ratio. Its first-principles status is therefore exactly the status of that gate, whose obstruction is isolated in AOG-Note [4] as an arithmetic orthogonality between Born–Infeld parity and the Galois-spin action. Without the explicit Lorentzian E_Π, the three-generation mass spectrum and the Yukawa sector cannot be derived.

2. *Normalisation of* $\mathcal{L}_Y$. The Yukawa Lagrangian $\mathcal{L}_Y$ is identified as arising from the trace $\mathrm{Tr}_{\mathcal{S}_\Pi}[E_\Pi^2]$ variation. Its integral normalisation and the derivation of the mass matrix from the spectral data of E_Π are open.

3. $[\text{H-color}]_{\text{pointwise}}$ *beyond the standard graph.* Inherited from Notes 1 and 3. $[\text{H-color}]_{\text{pointwise}}$ is proved on the standard Cayley graph (O31); its extension to general generating sets is the only residual, and is non-blocking for the quark bundle, which is already established at the pointwise level.

8 Open Deliverables

The derivation of the spinorial BI lift, formerly the first open deliverable, is now established as a theorem in Q14 [5]: the antilinear lift $J_\Pi = HK$ dresses the BI parity by the weight grading, making Theorems B and C unconditional. Two open deliverables remain.

Open deliverable 1: Explicit E_Π and the Yukawa sector. The mass spectrum of charged leptons and quarks, and the CKM/PMNS mixing matrices, require the explicit spectral construction of E_Π as an operator on the admissible Hilbert space. This requires integrating the projected Yukawa Lagrangian $\mathcal{L}_Y$ and extracting the mass matrix from the eigenvalues of E_Π^2. This is the primary open problem of the fermionic sector. The structural form of E_Π is now fixed as the Schur complement of the eliminated spinorial block [10]; the open part is its explicit Lorentzian block $1 - P(s)$ along the J_Π-odd modulus. The symmetric finite front is J_Π-equivariant and gives $u_{\text{fin}} = 0$, but the discrete chiral-frontier normalisation $\mathcal{N}_A$ survives the operator obstruction, since the residual reflection $R_b = \mathrm{FT}^2$ is blind to chirality ($[\gamma_5, R_b] = 0$) [9]. The split amplitude is therefore fixed at A4 by matching this discrete normalisation with the Lorentzian Born–Infeld curvature $\mu_\chi^2 = \partial_s^2 \mathcal{B}(0)$, neither ingredient reducing to the other:

$$|u| = \mathrm{A4Match}\big(\mathcal{N}_A,\, \mu_\chi^2\big)\,.$$

The operator side of this matching is no longer conditional: by [10] the Schur transversality is reduced to the non-vanishing of the projected commutator $[\partial_s P|_0, J_\Pi]\big|_{\text{span}(e_+,e_-)}$, with an exhaustive transverse/null dichotomy. In the Schur-transverse branch the A4 chiral polarisation is electric, hence $\mathcal{P}_{\mu\nu}\mathcal{P}^{\mu\nu} < 0$, $\mu_\chi^2 < 0$, and the split opens spontaneously [8]; the genus is determined. What remains open is the split value $|u|$ and the Yukawa sector downstream of u. On the corpus-derived real cascade the A4 companion resolves the boundary-versus-interior selection in favour of saturation: the transverse (spin-two) cubic response of $\mathcal{P} \cdot \mathcal{R}_\chi$ vanishes path-invariantly at $\gamma = 0$, so the derived amplitude mechanism is Born–Infeld saturation, and the sixth-order interior lock is doubly conditional — on a non-derived complex phase and a non-prescribed ordering — hence not concluded [8]. The value $|u|$ itself is not a parameter-free prediction: it is dictionary-bound through the chiral-frontier normalisation $\mathcal{N}_A$, the angular analogue of the radial scale $\kappa = 5/12$ of the exit-deficit dictionary [7, 5], whose split value $\varepsilon = 1/10$ is fixed by matching the derived three-level deficit structure to the external ADE ratio of a selected case [3]. The genuine first-principles front for the value is therefore upstream and structural, not a numerical amplitude: the ADE case selection and the level-to-generation map, the derivation of the projective-resolution growth and the cascade exponent β (only $\beta \le 1$ established [3, 2]), and the derivation of the transfer constant N_{casc} that fixes $\mathcal{N}_A$ [5]. What remains quantitative is therefore these upstream dictionary inputs, the integral normalisation of $\mathcal{L}_Y$, and generation mixing via the complex metaplectic phase.

Open deliverable 2: Full matter content in Lorentzian signature. The complete fermionic content $S_\Pi^{\mathrm{matter}} = (\mathcal{S}_\Pi \oplus (\mathcal{S}_\Pi \otimes V_{\mathrm{color}})) \otimes C^3_{\mathrm{gen}}$ in full Lorentzian signature, with explicit gauge couplings and three-generation Yukawa structure, is the downstream target. It requires the resolution of Deliverable 1 above, the colour sector being already established at the pointwise level (O31).

9 Conclusion

The fermionic matter sub-programme establishes that fermions, chirality, hypercharge, and three generations are not external additions to the Cosmochrony framework. They are the spinorial face of the admissible Weil module V_ρ, revealed through the Lorentzian complexification of its metaplectic structure.

The electroweak bundle $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ arises from the tensor functors of the admissible spinor bundle $\mathcal{S}_\Pi$ (Theorem A, unconditional). The $V - A$ chiral structure and hypercharge rigidity arise from the projective endomorphism E_Π (Theorem B, unconditional). Three generations arise from the spinorial multiplicity reading of $\sigma_c(n_3) = 3$ (Theorem C, unconditional). The colour-coupled quark sector follows by tensoring with V_{color}, unconditional at the pointwise level (O31).

The inter-generation splitting is localised: statically obstructed, living in a two-dimensional J_Π-odd lift sector, and vanishing on the symmetric finite front ($u_{\text{fin}} = 0$, J_Π-equivariant). Its operator obstruction is resolved, the residual reflection $R_b = \mathrm{FT}^2$ being blind to chirality ($[\gamma_5, R_b] = 0$), and its amplitude mechanism is the derived Born–Infeld saturation [8]. The remaining open components are the explicit Lorentzian E_Π; the split value, which is dictionary-bound and whose first-principles derivation is upstream and structural (the ADE case selection and level-to-generation map, the cascade exponent β, and the transfer constant N_{casc}); and the derivation of the Yukawa sector and mass spectrum.

References

[1] J. Beau. Admissible Non-Injective Transitions as the Primitive of Physical Description, 2026. Cosmochrony Foundation Paper "M", doi:10.5281/zenodo.19616329.

[2] J. Beau. Subdiffusive Valence Growth under Bounded Relational Flux: Structural Derivation of the Cascade Exponent β, 2026. Spectral admissibility sub-program "O4" paper, doi:10.5281/zenodo.19101472.

[3] J. Beau. Three Particle Generations and Mass Hierarchy from Spectral Stratigraphy, 2026. Spectral admissibility sub-program paper, doi:10.5281/zenodo.19040490.

[4] Jérôme Beau. Arithmetic Orthogonality of the ADE Case-Selection Gate, 2026. Fermionic matter / spectral-stratigraphy transversal note. doi:10.5281/zenodo.20693084.

[5] Jérôme Beau. Fermionic Matter and Chirality from Projective Dirac Admissibility, 2026. "Q14" paper, doi:10.5281/zenodo.20218409.

[6] Jérôme Beau. Gauge–Gravity Spectral Synthesis: Joint Einstein–Yang–Mills Equations, Born–Infeld Completion, and the Hierarchy Ratio from Projective Spectral Entropy, 2026. "Q13" paper, doi:10.5281/zenodo.20209859.

[7] Jérôme Beau. Reduction of the Angular Amplitude Observable: the J_3 Split as Accumulated Oriented Symplectic Area, 2026. doi:10.5281/zenodo.20601228.

[8] Jérôme Beau. The Born–Infeld Saturation Margin of the Chiral Modulus: Antisymmetric Structure and the Reduction of the Generation Split to a Lorentzian Genus, 2026. Fermionic matter sub-programme companion note. doi:10.5281/zenodo.20633931.

[9] Jérôme Beau. The Oriented Frontier Observable: a Recursive Q11 Diagnostic for the Angular Generation Split, 2026. doi:10.5281/zenodo.20601245.

[10] Jérôme Beau. The Projective Residue as a Schur Complement: Reduction of the Generation Split to the Chiral Asymmetry of Projection Locking, 2026. doi:10.5281/zenodo.20601040.

[11] Jérôme Beau. The Spectral Gravity Sub-Programme, 2026. Presentation Note 4, doi:10.5281/zenodo.20533351.

[12] Jérôme Beau. Three Admissible Directions and Three Spatial Dimensions: A Structural Bridge Candidate, 2026. "Q7" paper, doi:10.5281/zenodo.19802123.

[13] Jérôme Beau. Yang–Mills Dynamics from Projective Spectral Entropy: Vertical Heat-Kernel Variation on the Admissible Fibre, 2026. "Q12" paper, doi:10.5281/zenodo.20261952.